2.4 Nonclinical Overview

PrP-lowering divalent siRNA 2439-exNA

Other names: 2439-exNA, 2439-s4, PrP-siRNA

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List of Abbreviations

ASO	antisense oligonucleotide
CJD	Creutzfeldt-Jakob disease
CSF	cerebrospinal fluid
di-siRNA	divalent short interfering RNA
dpi	days post-infection
ED_{50}	median effective dose
ELISA	enzyme-linked immunosorbent assay
exNA	extended nucleic acid
FDA	Food and Drug Administration
FFI	Fatal familial insomnia
GSS	Gerstmann-Straussler-Scheinker disease
ICH	International Conference on Harmonization
PD	pharmacodynamic
PK	pharmacokinetic
PrP	prion protein
SAD	single ascending dose
siRNA	short interfering RNA

2.4 Nonclinical Overview

2.4.1 Overview of the Nonclinical Testing Strategy

PrP lowering divalent siRNA 2439-exNA (hereafter referred to as 2439-exNA) is a fully chemically modified small interfering RNA oligonucleotide that targets human *PRNP* messenger RNA, indicated for the treatment of human prion disease. Prion disease is a fatal, incurable neurodegenerative disease caused by misfolding of prion protein (PrP), encoded in humans by the gene *PRNP*[1]. Convergent lines of evidence implicate PrP as the drug target in this disease[2], and indicate that lowering PrP should be both safe[3–9] and effective[10–18].

PrP-lowering divalent siRNA 2439-exNA was discovered through screening of siRNAs in human cultured cells and in transgenic mice expressing human *PRNP* to determine the siRNA sequences and chemical backbones providing the deepest lowering of the *PRNP* RNA transcript and PrP protein. In human PrP transgenic mice dosed intracerebroventricular (ICV) injection, both the sequence and backbone of 2439-exNA were determined to be superior to other compounds tested (see section 4.2.1.1 Primary Pharmacodynamics). At high doses in mice (174 or 348 µg), suppression to as little as 28% or 17% residual PrP in whole brain hemisphere was observed, and activity was observed out to 4-6 months. In dose response, the median effective dose, or dose to achieve 50% lowering of *PRNP* mRNA), was determined to range from 5 to 18 µg across 6 brain regions evaluated. Based on CSF volume scaling (a factor of 3,250 from mouse to human), 18 µg may correspond to 59 mg in a human. Because 2439-exNA is not sequence matched for mouse *Prnp*, a surrogate compound targeting mouse *Prnp* was utilized to study the effects of divalent siRNA-mediated PrP lowering in a disease model, wild-type mice infected with the RML strain of prions. Survival time was increased 2.7-fold with early (pre-symptomatic) chronic divalent siRNA treatment, and survival was increased by 64% even with a single dose administered at a symptomatic disease stage.

The PK and tissue distribution of 2439-exNA was evaluated following single IT dose administrations in rats and beagle dogs. Neither species is pharmacologically responsive to inhibition of PrP by siRNA 2439-exNA, therefore the studies only assessed off-target effects, such as those related to oligonucleotide chemistry. The decision to use non-pharmacodynamic species was made in consultation with FDA at a Pre-IND meeting. An large body of evidence from decades of studies in animal models as well as using human genetic databases indicates that partial or even complete PrP lowering will not have adverse effects[2].

The nonclinical safety studies were consistent with the International Council for Harmonisation ICH M3(R2) *Nonclinical Safety Studies for the Conduct of Human Clinical Trials for Pharmaceuticals* (ICH 2009) and the pivotal studies were conducted in compliance with United States Food and Drug Administration (FDA) Good Laboratory Practice (GLP) Regulations (21 CFR Part 58). The same batch of 2439-exNA (Lot No. [redacted]) was used in the GLP IND-enabling studies and is representative of the clinical batch of material to be used in Phase 1 trials (see section 3.2.S).

siRNA 2439-exNA was evaluated for potential to induce or inhibit a series of CYPs using cultures of cryopreserved primary human hepatocytes, as an inhibitor of the human transporters P-gp, BCRP, OATP1B1, OATP1B3, OAT1, OAT3, OCT2, MATE1 and MATE2-K, and a substrate of human transporters P-gp and BCRP. Based on in vitro evaluations, 2439-exNA has low potential as a substrate or inhibitor of metabolizing enzymes and membrane transporters.

The completed toxicology studies include single dose studies in rats and dogs, with necropsies at 24 hours and 28-days. The intrathecal (IT) route of administration was used for the nonclinical safety studies because it is the proposed clinical route. No adverse toxicity was identified in the completed siRNA 2439-exNA GLP toxicology studies in rats and dogs. The NOAEL in rats was 3.0 mg and the NOAEL in dogs was 200 mg, the highest dose tested in each species.

Safety pharmacology endpoints (i.e., respiratory, cardiovascular and neurobehavioral endpoints) were incorporated into the dog GLP study, there were no effects in these evaluations. Genotoxicity was evaluated bacterial reverse mutation and in vitro micronucleus assays; siRNA 2439-exNA was negative for genotoxicity.

The results from the nonclinical pharmacology, pharmacokinetic and toxicology studies conducted with siRNA 2439-exNA fully support the initiation of the proposed Phase 1 clinical trials in patients with confirmed prion disease based on the substantial safety margins in the GLP toxicology studies (26.7-fold to 40.0-fold) over the FIH starting dose.

2.4.2 Pharmacology

Therapeutic hypothesis of PrP lowering.

Decades of evidence from animal models as well as humans support the safety of PrP lowering[2]. PrP knockout mice were first engineered in 1992, and initially defied any efforts to identify a phenotype other than total resistance to prion disease[3]. Additional lines of PrP knockout mice using alternative gene disruption strategies were later engineered, for a total of 8 extant PrP knockout lines[19]. Because PrP is a highly studied gene, a large number of scientific papers have been published on different observations in PrP knockout mice, however, a large majority of assertions of PrP knockout phenotypes have never been replicated, and many have been refuted or ascribed to genetic confounders[20]. Following 18 years of intensive study, just one knockout phenotype emerged as reproducible[6]. Aged homozygous PrP knockout mice develop a mild peripheral neuropathy associated with insufficient maintenance of myelin on peripheral nerves[6]. This phenotype is not observed in the central nervous system, but only the peripheral nervous system; not in heterozygous knockout animals, but only in homozygotes; and not in young but only in aged animals[6]. The phenotype has replicated in PrP knockout goats[21]. While the change is histologically evident when sciatic nerves are sectioned and stained, observing a phenotypic impact of this neuropathy has proven challenging. Delays in hot plate response and reduced grip strength were observed in aged homozygous knockout mice on a Balb/c genetic background but not on a C57BL/6 genetic background[6].

PrP lowering has already been well-established as a disease-modifying therapeutic strategy in prion disease through both genetic experiments and proof-of-concept experiments using antisense oligonucleotides[2,18]. In experiments described in detail in 4.2.1.1. Primary Pharmacodynamics and summarized briefly below, we further validated this therapeutic hypothesis by demonstrating that PrP lowering by a tool compound divalent siRNA targeting the mouse *Prnp* gene is effective at extending survival time in prion-infected mice.

Mechanism of action of 2439-exNA.

siRNAs are loaded into the RNA-induced silencing complex (RISC) and bind their complementary RNA targets through Watson-Crick base pairing and induce cleavage of those targets, lowering steady state levels of target RNA.

PrP-lowering divalent siRNA 2439-exNA consists of two covalently linked siRNA sense strands, each hybridized to an siRNA antisense strand. All strands are fully chemically modified to promote cellular uptake and exonuclease resistance, increasing mechanism of action. Like all oligonucleotide drugs, divalent siRNA are taken up into endosomes where they form a slow-release depot that is gradually released into cytosol, resulting in sustained pharmacologic activity over the course of months. 2439-exNA is designed to be complementary to the human *PRNP* RNA, encoding the prion protein (PrP), which is the cause of both initiation and progression in all cases of prion disease regardless of subtype or etiology. 2439-exNA is intended to be dosed into cerebrospinal fluid (CSF). This would be accomplished by an intrathecal (IT) route in human and accordingly IT was used as the delivery route in our GLP toxicology studies. In proof-of-concept studies in mice, given the difficulty of IT delivery in this smaller animal, we utilized intracerebroventricular (ICV) delivery.

2439-exNA was identified as a top candidate out of 84 siRNA sequences screened in cultured cells for *PRNP* RNA lowering in human cells. Using transgenic mice harboring the entire human *PRNP* gene, including non-coding regions, we tested 13 sequence/chemical scaffold combinations in vivo, using whole brain hemisphere PrP protein as a readout. 2439-exNA was identified as the top performing compound in this screen. Further characterization studies of the median effective dose (ED50) and durability of 2439-exNA in this transgenic mouse model indicate that 2439-exNA is highly potent at lowering PrP. It has no predicted off-targets with just 1 mismatched base in the human reference transcriptome, supporting its specificity. GLP toxicology studies have indicated a non-adverse profile at single doses up to 200 mg in dog and 3 mg in rat.

2.4.2.1 Primary Pharmacodynamics

Proof of concept in a disease model. We identified a tool compound with a sequence specific to the mouse *Prnp* gene and used it to establish the efficacy of PrP-lowering divalent siRNA therapy in prion-infected mice. We utilized a standard disease model — wild-type mice inoculated intracerebrally with RML prions — and used intracerebroventricular (ICV) delivery of divalent siRNA into CSF. Treatment every 4 months, beginning at a disease stage with neuropathology but before frank symptoms, with a dose that lowers PrP to 49% residual, yielded a 2.7-fold increase in survival time (from 165 to 442 days on the median). Treatment with a single dose upon the onset of symptomatic disease indicated by weight loss yielded a 64% increase in survival time (from 164 to 270 days on the median).

Identification of drug candidate. We screened 84 siRNA sequences in human U251-MG glioblastoma cells to identify sequences that potently lower human *PRNP.* Using transgenic mice harboring the entire human *PRNP* gene, including non-coding regions, we tested 13 sequence/chemical scaffold combinations in vivo by intracerebroventricular (ICV) delivery, using whole brain hemisphere PrP protein as a readout. 2439-exNA was identified as the top performing compound in this screen. We further compared chemical scaffold configurations, varying the 2' modifications and internucleotide linkage pattern as well as fixed versus matched 3' antisense tails. Through these studies we determined that a chemical scaffold denoted s4, incorporating extended nucleic acid and a 3′ antisense tail unmatched to the RNA target, yields superior in vivo potency. Collectively of these studies nominated PrP-lowering divalent siRNA 2439-exNA, as our drug candidate. In dose-response experiments in transgenic mice expressing human PrP, we find that the median effective dose (ED_{50}) — the dose resulting in 50% lowering of *PRNP* transcript by RT-qPCR — of this drug candidate ranges from 5 µg to 18 µg across the 6 mouse brain regions tested. A

single dose at 52 µg in mouse resulted in 49% residual PrP in whole brain hemisphere, and higher doses of 174 µg and 348 µg yielded 29% and 17% residual PrP after a single dose, with durability out to 4-6 months.

2.4.2.2 Secondary Pharmacodynamics

We performed an in silico assessment of the potential hybridization-mediated off-target effects of 2439-exNA, described in detail in 4.2.1.2 Secondary Pharmacology and summarized briefly here.

The 16 bases from 2-17 are most critical for determining activity against the target RNA. This critical region of 2439-exNA is GAAUACUCACAAAGUG. The reverse complement of this, CACTTTGTGAGTATTC, is the target sequence present in the coding (sense) strand of *PRNP* DNA corresponding to the 3'UTR of the mRNA. This sequence was searched against the Homo sapiens RefSeq Select mRNAs using NCBI Nucleotide BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi). This search confirmed *PRNP* as the sole gene with 0 mismatches to 2439-exNA. No genes had exactly 1 mismatch. 2 genes had 2 mismatches each: *FURIN* and *ZKSCAN3*. All other genes had at least 3 mismatches.

Overall, evidence in vivo from the CNS indicates that transcripts with ≥2 mismatches are unlikely to experience any off target lowering[22,23]. In the human transcriptome, all non-*PRNP* genes have at least 2 mismatches to 2439-exNA, thus minimizing the risk of unwanted off-target effects.

2.4.2.3 Safety Pharmacology

No stand-alone, safety pharmacology studies have been performed with 2439-exNA. However, appropriate safety pharmacology endpoints e.g., cardiovascular, respiratory functions, and blood pressure measurements were included in the single dose GLP dog toxicology study, and neurobehavioral function (functional observation battery) was included in the single dose GLP rat toxicology study. No test article related effects were observed for the safety pharmacology endpoints measured.

2.4.2.4 Pharmacodynamic Drug Interactions

No studies were conducted.

2.4.3 Pharmacokinetics

Single dose 2439-exNA toxicokinetic (TK) evaluations were conducted in the rat and dog. 2439-exNA was administered via IT injection in GLP toxicology studies, because this is the proposed clinical route. In addition to the plasma toxicokinetic evaluation, CSF and tissue concentrations were also measured.

GLP toxicology studies used the same GMP batch of Drug Substance as was used in clinical Drug Product formulation. Moreover, in order to best represent the clinical Drug Product, the GLP toxicology studies also used sodium chloride, phosphate buffer, and magnesium chloride as excipients. The molar ratio of magnesium ions to 2439-exNA was 25:1 in the GLP toxicology studies, just as in the clinical formulation.

At the time of this IND submission, tissue and CSF concentrations are available for dogs, but analysis of rat samples are ongoing.

2.4.3.1 Methods of Analysis

The TK plasma assays as well as bioanalytical evaluation of 2439-exNA in CSF and tissue samples employed a hybridization-based Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) method coupled to a fluorescence detector.

2.4.3.2 Absorption

Across single-dose studies, 2439-exNA demonstrated a profile which is consistent with IT administration of siRNA with evidence of uptake into the brain and liver tissue, and excretion via the kidneys. Following IT dose administration, the plasma concentration-time profiles for siRNA 2439-exNA males and females showed an apparent monophasic decline. T_{max} occurred at either the first or second target collection time of 0.5 or 4.00 hr in dogs and at the first collection time of 0.5 hours in rats.

Across single-dose studies, 2439-exNA demonstrated a profile which is consistent with IT administration of siRNA molecules with evidence of uptake into the brain and liver tissue, and excretion via the kidneys. Following IT dose administration, the plasma concentration-time profiles for PrP lowering divalent siRNA 2439-exNA males and females showed an apparent monophasic decline. T_{max} occurred at either the first or second target collection time of 0.5 or 4.00 hr in dogs and at the first collection time of 0.5 hours in rats.

2.4.3.3 Distribution

2439-exNA was detected in the CSF for most dogs and all rats on Day 2 (~24 hours) following 2439-exNA administration, although the concentrations were highly variable. However, even in the subset of dogs where 2439-exNA was not detected in CSF, exposures in spinal cord and brain confirmed that 2439-exNA had been delivered to the intrathecal space.

Evaluation of 2439-exNA tissues levels in dogs showed that the highest concentration 2439-exNA was present in the liver. 2439-exNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. 2439-exNA was observed in the brain, with frontal cortex, hippocampus, and cerebellar cortex generally containing higher concentrations than the thalamus, putamen, and caudate. 2439-exNA was also observed in the kidney. Generally, 2439-exNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2. The concentration of 2439-exNA demonstrated a dose proportional trend.

2.4.3.4 Metabolism

No metabolism studies were conducted or planned with 2439-exNA.

2.4.3.5 Excretion

No excretion studies were conducted or planned with 2439-exNA.

2.4.3.6 Pharmacokinetic Drug Interactions

Based on in vitro evaluations, 2439-exNA has low potential as a substrate or inhibitor of metabolizing enzymes and membrane transporters.

2.4.4 Toxicology

The nonclinical toxicology studies for 2439-exNA were designed to enable Phase 1 SAD studies. The completed nonclinical toxicology studies are shown in Table 1.

Table 1: Completed 2439-exNA Nonclinical Safety Studies

Species / Test System	Study Description Duration (weeks)	Route of Administration	GLP Status	Study Number
Studies completed for original IND submission				
Rat	Single dose study with 4-week recovery	IT	GLP	A-703-01
Dog	Single dose study with 4-week recovery	IT	GLP	A-703-02
Dog Dose Range Finding	Maximum tolerated dose study in dogs	IT	Non-GLP	A-703-03
Bacterial reverse mutation and in	*Salmonella typhimurium* (TA1535, TA1537, TA98, and TA100) and *Escherichia coli* (WP2 trp uvrA) strains	N/A	GLP	56772
In vitro micronucleus assays	CHO-K1 Cells	N/A	GLP	56773

The IT route of administration was used for the nonclinical toxicology studies because it is the proposed clinical route. Single dose studies, with necropsies at ~24 hours and 28 days, were conducted as the SAD clinical study will include only a single dose administration. A non-GLP maximum tolerated dose study in dogs was performed with a high dose of 200 mg, to identify the maximum dose for the subsequent GLP study in dogs. The highest doses tested in dog and rat GLP studies were 200 mg and 3.0 mg, respectively.

Safety pharmacology endpoints (i.e., respiratory, cardiovascular and neurobehavioral endpoints) were incorporated into the dog GLP study. Genotoxicity was evaluated bacterial reverse mutation and in vitro micronucleus assays.

The batch of 2439-exNA (Lot No. [redacted]) that was used in the GLP IND-enabling studies was the same that will be used in Phase 1 trials. The nonclinical safety studies were consistent with the International Council for Harmonisation ICH M3(R2*) Nonclinical Safety Studies for the Conduct of Human Clinical Trials for Pharmaceuticals* (ICH 2009) and the pivotal studies were conducted in compliance with United States Food and Drug Administration (FDA) Good Laboratory Practice (GLP) Regulations (21 CFR Part 58).

The design and findings of the Toxicology studies are further summarized below and in 2.6.6.

2.4.4.1 Single-Dose Toxicity

2.4.4.1.1 Maximum Tolerated Intrathecal Dose Study in Dogs (Study A-703-03)

The purpose of this study was to determine the maximum tolerated dose of 2439-exNA administered once via intrathecal injection to Beagle dogs. The highest dose was selected to be 200 mg in dog (16 mL CSF), in order to provide an approximately 8-fold margin, based on CSF volume scaling[24], over the max feasible clinical dose of 200 mg in human (130 mL CSF). One dog per sex was administered 200 mg 2439-exNA via intrathecal slow bolus over a duration of 5 minutes and observed for 4 days following dose administration. There were no clinical observations attributed to siRNA 2439-exNA following administration of the high dose of 200 mg. With no adverse effects following administration at 200 mg, additional dose levels were not evaluated. A dose of 200 mg was considered well tolerated in Beagle dogs.

2.4.4.1.2 A Single Intrathecal Dose Toxicology and Toxicokinetic Study in Dogs with a 28-Day Recovery Period (Study A-703-02)

The purpose of this study was to evaluate the potential toxicity and toxicokinetics of 2439-exNA following a single intrathecal injection to Beagle dogs. The reversibility of any toxicological effects were evaluated following a 28-day recovery period.

Forty-eight (24 males/24 females) Beagle dogs were assigned to one of four dose groups: Group 1, Vehicle Control (0 mg), Group 2, Low (20 mg), Group 3, Mid (60 mg), and Group 4, High (200 mg) 2439-exNA. On Day 1, the animals were anesthetized and administered the appropriate test or control material via intrathecal injection administered over 3-5 minutes. On Day 2 four animals per sex per group and on Day 29 two animals per sex per group were humanely terminated and a necropsy with select tissue collection was performed.

There was no mortality observed; all animals survived to scheduled termination. Clinical observations observed post dose administration on Day 1 consisted of decreased activity, lethargy, excessive salivation, impaired limb use/limb weakness, and these observations resolved by Day 2 in the males but remained on Day 2 in the female animals and were attributed to the anesthesia and intrathecal administration procedure. There were no effects on body weights, food consumption, neurobehavioral observations, ophthalmic observations, respiratory rates, ECG waveforms, or cardiovascular parameters, or clinical pathology measurements attributed to the administration of 2439-exNA.

There were no test article related macroscopic necropsy observations. Microscopic test article related findings were limited to minimal focal to multifocal gliosis in the brain and spinal cord of one 60 mg 2439-exNA and four 200 mg 2439-exNA Core animals. Gliosis involved increases in indeterminate glial cells and therefore was non-specific evidence of inflammation or cell damage. Gliosis can manifest within 24 hours of injury and may remain for days or weeks, depending on the severity and chronicity of the inciting insult. Only one male 200 mg 2439-exNA animal bore evidence of proximal cause of gliosis, focal necrosis in the spinal cord gray matter at the intrathecal injection site, with adjacent minimal gliosis in the cerebrum. For the remaining four animals, there is no indication of cell damage to incite gliosis, leaving test article as the presumed cause (lacking similar findings in the Vehicle Control, 20 mg 2439-exNA dose, or recovery animals). These minimal findings were considered non-adverse as there was no changes in the blood nor correlated clinical signs.

In conclusion, there were no test article related adverse findings at 20 mg, and there were no adverse test article related findings at the 60 and 200 mg 2439-exNA dose levels. Non-adverse test article related findings in the 60 and 200 mg 2439-exNA groups were limited to gliosis. Therefore, 200 mg siRNA 2439-exNA is considered the no observable adverse effect level (NOAEL).

2.4.4.1.3 A Single Intrathecal Dose Toxicology and Toxicokinetic Study in Rats with a 28-Day Recovery Period (Study A-703-03)

The purpose of this study was to evaluate the potential toxicity and toxicokinetics of 2439-exNA following a single intrathecal injection to Sprague Dawley rats. The reversibility of any toxicological effects was evaluated following a 28-day recovery period.

One hundred sixty-two (81 males/81 females) Sprague Dawley rats were assigned to one of four dose groups: Group 1, Vehicle Control (0 mg), Group 2-Low (0.3 mg), Group 3-Mid (1.0 mg), and Group 4-High (3.0 mg) 2439-exNA. On Day 1, the animals were administered the appropriate test or control article via intrathecal bolus injection. On Days 2 and 29, core and recovery animals, respectively, were humanely terminated and a necropsy with select tissue collection was performed.

There was no mortality or clinical observations, changes in body weights, body weight gains, food consumption or ocular observations attributed to the administration of 2439-exNA. There were no test article-related changes in hematology, coagulation, serum chemistry, or urine parameters examined. Sprague Dawley rats given up to 3.0 mg 2439-exNA by intrathecal bolus injection on Day 1 and terminated on Day 2 had no gross or microscopic findings related to the test article. Similarly, recovery animals terminated on Day 29 had no test article-related findings.

Overall, there were no observed adverse effects related to intrathecal administration of 2439-exNA to Sprague Dawley rats across the dose range tested of 0.3 to 3.0 mg. The No Observed Adverse Effect Level (NOAEL) was determined to be 3.0 mg.

2.4.4.2 Genotoxicity

2439-exNA was evaluated for genotoxic potential in alignment with Option 1 of ICH S2 (R1) (2011). The studies conducted were the in vitro bacterial reverse mutation assay (GLP) and the *vitro* micronucleus test with the Chinese hamster ovary derived CHO-K1 cells (GLP). The highest concentrations of 2439-exNA used for the in vitro assays, 5000 mcg/plate, were the maximum dose level recommended per ICH S2A(R1).

2.4.4.2.1 In vitro studies

2.4.4.2.1.1 Bacterial reverse mutation assay

The objective of the study was to evaluate the ability of 2439-exNA to induce reverse mutations in *Salmonella* and *E. coli* strains in the presence or absence of an exogenous mammalian metabolic activation system containing microsomal enzymes (S9). The study was conducted according to the ICH S2 (R1) and OECD-471 test guidelines.

2439-exNA did not induce an increase in the revertant frequencies (≥ 2-fold) as compared to that of the vehicle/negative controls, for all concentrations tested under both non-activated as well as S9-activated metabolic conditions. In conclusion, under the conditions of this study, 2439-exNA was devoid of any

mutagenic potential in the bacterial reverse mutation test (Ames test), when tested on *S. typhimurium* strains TA1535, TA1537, TA98, TA100, and on the *E. coli* strain WP2 trp uvrA, with and without metabolic activation (±S9), at dose levels up to the Guideline limit dose of 5000 µg/plate.

2.4.4.2.1.2 Micronucleus assay

The objective of this study was to evaluate the cytogenetic potential of 2439-exNA, using the *in vitro* micronucleus test with the Chinese hamster ovary derived CHO-K1 cells. The study was conducted according to the ICH S2 (R1) and OECD-487 test guidelines.

Cultures treated with 2439-exNA did not show any statistically significant increase in the incidence of micronuclei. The numbers of micronuclei were comparable with the concurrent vehicle/negative control and were within or close to the historical control range. There was no significant increase in hypodiploid cells, indicative of numerical aberrations, in either the short-duration or extended treatments with 2439-exNA, in the absence or presence of S9.

In conclusion, 2439-exNA was considered negative for clastogenic or aneugenic activity in the *in vitro* micronucleus assay based on the lack of increased micronuclei or hypodiploid frequencies in CHO-K1 cells following 4-hour incubation with and without metabolic activation and 26-hour incubation without metabolic activation at dose levels up to the Guideline limit dose of 500 µg/mL.

2.4.4.4 Other toxicity studies

No other toxicity studies have been performed.

2.4.5 Integrated Overview and Conclusions

The toxicology studies conducted with 2439-exNA adequately characterized the nonclinical safety of 2439-exNA. Single dose GLP nonclinical toxicology studies were conducted in two species, the rat and dog. The studies used intrathecal administration, which is the intended route of administration for patients. The toxicology assessment did not identify adverse toxicity following a single SC dose of 2439-exNA of up to 3.0 mg in rat and 200 mg in the dog GLP toxicology studies.

siRNA 2439-exNA was negative in both the bacterial reverse mutation assay and in the chromosomal aberration assay in cultured Chinese hamster lung fibroblast with or without metabolic activation.

Single dose 2439-exNA toxicokinetic (TK) evaluations were conducted in the rat and dog. 2439-exNA was administered via IT injection in GLP toxicology studies. In addition to tissue and CSF concentrations measured in these studies, plasma toxicokinetic evaluation was also conducted. At the time of this IND submission, tissue and CSF concentrations are available for dogs, but analysis of rat samples are ongoing.

Across single-dose studies, 2439-exNA demonstrated a profile which is consistent with IT administration of siRNA molecules with evidence of uptake into the brain and liver tissue, and excretion via the kidneys. Following IT dose administration, the plasma concentration-time profiles for PrP lowering divalent siRNA 2439-exNA males and females showed an apparent monophasic decline. T_{max} occurred at either the first or second target collection time of 0.5 or 4.00 hr in dogs and at the first collection time of 0.5 hours in rats.

2439-exNA was detected in the CSF for most dogs and all rats on Day 2 (~24 hours) following 2439-exNA administration, although the concentrations were highly variable. However, even in the subset of dogs

where 2439-exNA was not detected in CSF, exposures in spinal cord and brain confirmed that 2439-exNA was delivered to the intrathecal space.

Evaluation of 2439-exNA tissues levels in dogs showed that the highest concentration 2439-exNA was present in the liver. 2439-exNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. 2439-exNA was observed in the brain, with frontal cortex, hippocampus, and cerebellar cortex generally containing higher concentrations than the thalamus, putamen, and caudate. 2439-exNA was also observed in the kidney. Generally, siRNA 2439-exNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2. The concentration of siRNA 2439-exNA demonstrated a dose proportional trend.

Based on in vitro evaluations, 2439-exNA has low potential as a substrate or inhibitor of metabolizing enzymes and membrane transporters.

Based on the nonclinical safety data, the proposed clinical starting dose is 50 mg, which provides a >26-fold margin when calculated by local concentration at the injection site or by CSF volume scaling (Table 2) . The highest proposed clinical dose, 200 mg, has a margin of >6-fold.

Table 2. Proposed dose levels compared to toxicology data.

species	CSF volume (mL)	dose volume (mL)	NOAEL (mg)	proposed human doses (mg)	local concentration (mg/mL)	CSF concentration (mg/mL)	margin by local vs. rat	margin by local vs. dog	margin by CSF vs. rat	margin by CSF vs. dog
rat	0.25	0.03	3	–	100.0	12.0				
dog	16	3	200	–	66.7	12.5				
human	130	20	–	50	2.5	0.4	40.0	26.7	31.2	32.5
				100	5.0	0.8	20.0	13.3	15.6	16.3
				200	10.0	1.5	10.0	6.7	7.8	8.1

Safety margins to the proposed dose levels in humans to the NOAELs for rat and dog, using local injection concentration and CSF volumes (Emami et al[24,25])

2.4.6 Planned nonclinical studies

We are planning repeat-dose GLP toxicology studies in dog and rat to support repeat dosing in future human trials. In section 1.12.4 we pose a question the agency about whether our proposed design for these studies is acceptable.

2.4.7 Literature References

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